M2: Initial Implementation

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***Abstract*—This data mining project applies clustering techniques to the MovieLens dataset to discover natural patterns in how audiences receive and engage with films. Through comprehensive preprocessing including handling missing values, outlier treatment, standardization, etc., a rich feature set was engineered capturing movie ratings, popularity, genre composition, and user-generated metadata. K-means clustering with k = 4 revealed a hierarchical structure in cinema: the "Long Tail" of average films, "Mid-"All-Time Classics". These findings demonstrate that movie success follows predictable patterns, with implications for content strategy and audience understanding.**

I. Introduction

A. Background

In the age of streaming platforms and digital content libraries, understanding how audiences engage with films has become increasingly valuable. The MovieLens dataset, containing over 200,000 user ratings of over 85,000 movies, offers a rich source for discovering patterns in film reception. This project applies data mining techniques to uncover natural groupings of movies based on how they are rated and viewed, moving beyond simple genre classifications to reveal deeper structures in cinematic culture.

B. Discovery Questions

1. What natural group of movies exist based on user ratings, genres, and tags?
2. Are there distinct types of users based on their rating activity, genre preferences, and tagging behavior?
3. Are there any seasonal trends or binge-like rating behavior among users?

II. Exploratory Data Analysis

The purpose of this EDA was to structure, distribution properties, and the quality of the movie dataset prior to applying any data mining techniques. This step helps identify patterns, potential data quality issues, and relationships between variables that might influence the discovery of natural movie groupings.

A. Univariate Analysis - Numerical and Categorical Variables

Distribution of Raw User Ratings: The ratings range from 0.5 to 5.0 which increase at increments of 0.5. There appears to be a strong positive bias as most frequent rating is 4.0, followed by 3.0 and 5.0. Half-star increments are used less frequently than whole stars, suggesting users tend to think in whole numbers when providing feedback.

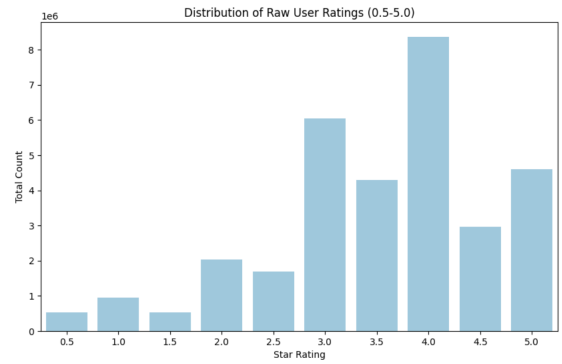

Fig. 1. Ratings Counts

Distribution of Raw Genres: The raw dataset includes numerous genres; however, Drama and Comedy tend to represent around 40% of the library. Moreover, the suspicious category of "no genre listed" was identified which must be properly handled in cleaned during prepossessing before applying any data mining techniques.

Genre Frequency: Drama and Comedy are the most prevalent genres, while Documentary and Animation appear less frequently. This imbalance suggests that clustering may be influenced more strongly by dominant genres unless dimensionality reduction is applied.

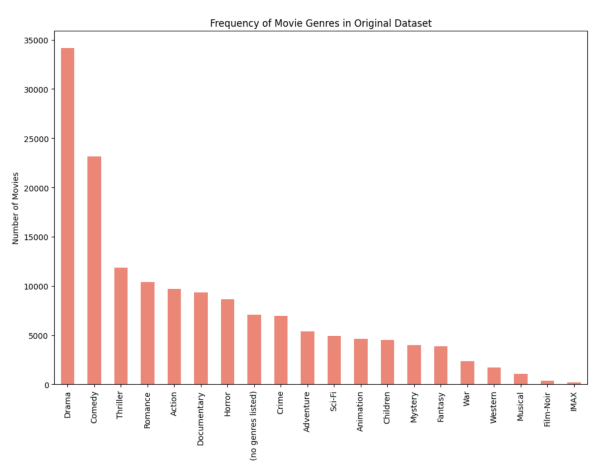

Fig. 2. Genres

B. Bivariate and Multivariate Analysis

Rating Distribution by Genre: When comparing top genres in the raw state, the median rating remains remarkably stable. This reveals that genre is not necessarily a good indicator of quality when it comes to movies. Therefore, this justifies the use of Clustering using K-Means as a viable option including "Popularity" and "Tags" as additional dimensions.

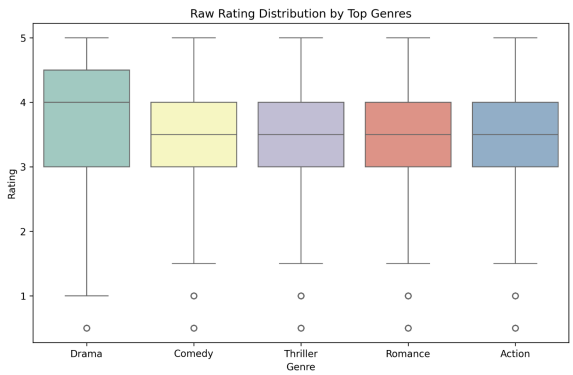

Fig. 3. Genre Distribution

III. Data Preprocessing

Preprocessing is a critical step in data analysis to ensure high quality results by cleaning the data, handling missing values, engineering useful features, and normalizing numeric variables before making it suitable for K-Means Clustering.
The preprocessing pipeline merged information from three MovieLens datasets: ratings, movies, and tags, and created a feature set suitable for pattern discovery.

A. Missing Values

Overall, the MovieLens dataset is remarkably clean, but the primary area that showed signs of missingness was the user-generated tags which was noticed during the integration of movie and tag data. Instead of removing these records, missing tag values were replaced with 0 in the metadata_richness feature. This approach reflects the semantic meaning of the data: a value of zero indicates that a movie simply has no user-generated tags rather than just removing them. Removing these rows would have introduced a "popularity bias" by only keeping movies that were heavily discussed.

```
tag_counts = tags.groupby('movieId')['tag'].count().reset_index(name='metadata_richness')
movie_df = movies.merge(movie_stats, on='movieId', how='inner')
movie_df = movie_df.merge(tag_counts, on='movieId', how='left').fillna(0)
```

B. Outliers

In this dataset, extreme values were primarily observed in the number of ratings per movie, where a small number of highly popular movies had thousands of ratings while many others had relatively few. To prevent these outliers from dominating the Euclidean distance in K-Means, we applied a

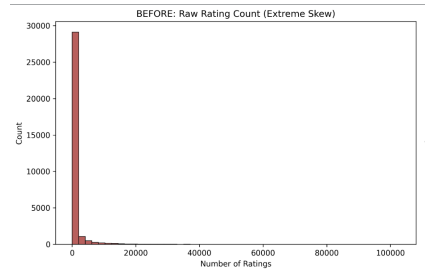

Fig. 4. Outlier Before

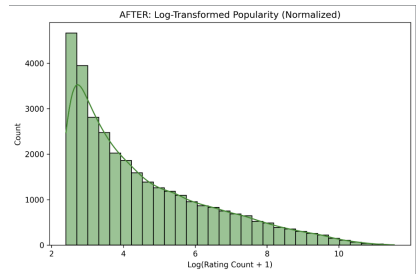

Fig. 5. Outlier After

Log Transformation ($log(x+1)$) to create the popularity_log feature.
This Log transformation preserves the ordering of popularity while compressing the scale, preventing blockbuster movies from dominating distance calculations in clustering.

In addition, the movie titles in movies.csv contained release years embedded in the title strings. However, some titles may not include a valid year, so a validation check was implemented when extracting the year.

```
def extract_year(title):
    year = re.search(r'\((\d{4})\)', title)
    return int(year.group(1)) if year else 0
movie_df['release_year'] = movie_df['title'].apply(extract_year)
```

C. Data Cleaning

Several data cleaning steps were performed to remove invalid data and ensure consistency.

First, duplicate records were removed from the ratings, movies, and tags datasets to prevent redundant entries from biasing the analysis.

```
ratings = pd.read_csv("data/ratings.csv").drop_duplicates()
movies = pd.read_csv("data/movies.csv").drop_duplicates()
tags = pd.read_csv("data/tags.csv").drop_duplicates()
```

Second, the original dataset stored timestamp fields as Unix epoch values (seconds since midnight (UTC) January 1, 1970.) which were converted into human readable date-time formats. This conversion enabled temporal feature engineering for user behavior analysis and temporal feature extraction to use if needed for future use.

```
ratings['timestamp_dt'] = pd.to_datetime(ratings['timestamp'], unit='s')
tags['timestamp_dt'] = pd.to_datetime(tags['timestamp'], unit='s')
```

Lastly, a threshold of minimum 10 ratings was added to ensure more reliable aggregate statistics, since movies with very few ratings could possibly produce unstable average scores and introducing noise.

```
movie_stats = movie_stats[movie_stats['rating_count'] >= 10]
```

D. Feature Engineering & Normalization

Feature engineering was performed at both the movie level and the user level to construct informative attributes that better capture behavioral patterns.

- **Date Extraction**: Regular expressions were used to extract release_year from the movie title strings and extract month from the rating timestamps to investigate seasonal trends.
- **Binge Behavior**: max_daily_ratings were derived by aggregating ratings per user per calendar day. This was done to identify "Power Raters" who rate dozens of movies in a single burst.

```
user_daily = ratings.groupby(['userId', ratings['timestamp_dt'].dt.date]).size().reset_index(name='daily_cnt')
binge_feature = user_daily.groupby('userId')['daily_cnt'].max().reset_index(name='max_daily_ratings')
```

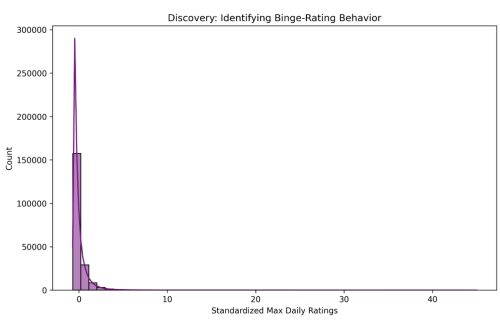

Fig. 6. User Binge Behavior

- **Seasonal Activity**: User rating timestamps were used to determine each user's most active month (peak_month). This feature captures seasonal engagement patterns.

```
ratings['month'] = ratings['timestamp_dt'].dt.month
user_seasonality = ratings.groupby('userId')['month'].agg(lambda x: x.value_counts().index[0]).reset_index(name='peak_month')
```

- **Tagging Behavior**: User interaction with the tagging system was quantified by total_tags_given. This made it possible represent how actively a user contributes metadata to the system.

```
user_tags = tags.groupby('userId')['tag'].count().reset_index(name='total_tags_given')
```

- **Rating Behavior Statistics**: Additional summary statistics were computed for each user. Average rating given, total number of ratings submitted, and rating standard deviation. These metrics help in capturing both user activity levels and rating tendencies.

```
user_stats = ratings.groupby('userId').agg(
    avg_user_rating=('rating', 'mean'),
    total_ratings_given=('rating', 'count'),
    rating_standard_dev=('rating', 'std')
).reset_index().fillna(0)
```

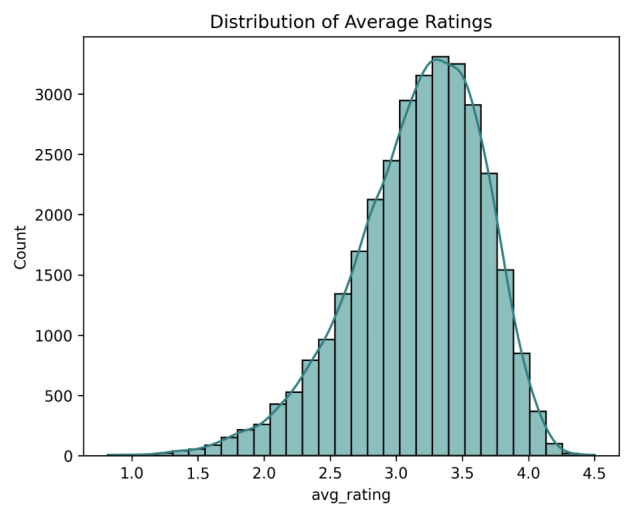

Fig. 7. Rating Distribution

- **Z-Score Normalization**: There was a massive scale disparity between the average ratings and metadata richness. Therefore, StandardScaler() was applied to all numeric features to have a mean of 0 and standard deviation of 1. This made it so the numeric features would share a common scale, enabling fair contribution to clustering algorithms.

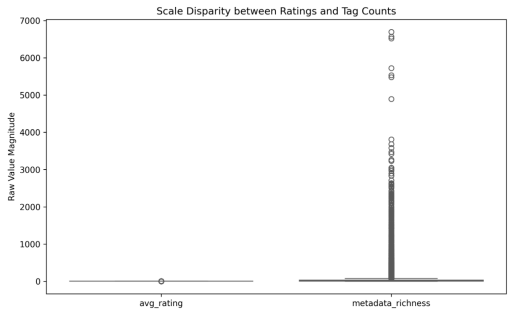

Fig. 8. Scale Disparity

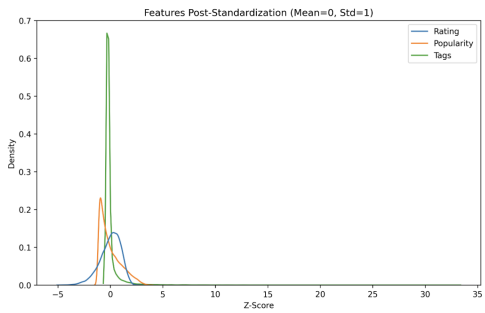

Fig. 9. Standardized Features

- **Feature Relationship Analysis**: To better understand relationships between engineered features, a correlation heatmap was generated for the primary movie-level variables: average rating, popularity (log-transformed rating count), metadata richness, and release year.

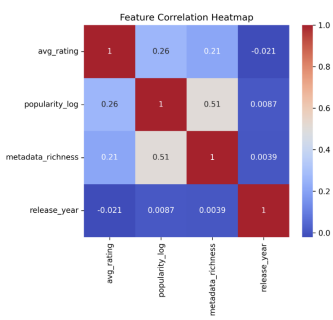

Fig. 10. Feature Heatmap

The heatmap reveals that popularity_log and metadata_richness exhibit a moderate positive correlation, suggesting that movies receiving more ratings also tend to accumulate more user-generated tags. This relationship is intuitive since highly visible movies attract both ratings and user discussion.

Average rating shows weak correlation with the other variables, indicating that the perceived quality of a movie is relatively independent of its popularity or tagging activity. This is beneficial for clustering because it allows rating-based preferences to influence grouping independently of popularity effects.

Release year also demonstrates low correlation with most variables, meaning that temporal information provides an additional dimension that may help differentiate clusters of older versus newer films.

Overall, the correlation analysis confirms that the engineered features capture different behavioral and descriptive aspects of movies, making them appropriate inputs for clustering algorithms such as K-Means.

E. Result of Preprocessing

After applying these preprocessing techniques, two structured datasets were produced.

- **Movie Dataset**: This dataset (preprocessed_movies.csv) contains all the engineered features required to perform the K-Means Clustering algorithm.
- **User Dataset**: This dataset (preprocessed_users.csv) contains all the engineered features required to analyze any user behavior or activity patterns.

IV. Mining Technique Application

This section presents the application of K-means clustering to discover natural groupings of movies based on their rating patterns, popularity, genre composition, and user-generated metadata. The goal was to identify whether distinct archetypes of films emerge from the data, moving beyond simple genre classifications to reveal deeper patterns in how audiences receive and engage with different types of movies addressing Discovery Question 1.

A. Feature Preparation

- **Numeric Features**: Average rating, popularity (log-transformed), and metadata richness (tag counts). These were previously scaled using StandardScaler to ensure a mean of 0 and standard deviation of 1.
- **Categorical Features**: 18+ genre categories, transformed via one-hot encoding.

Feature Category	Feature	Preparation
Rating Metrics	avg_rating	Standardized to mean=0, std=1 (from preprocessing)
Popularity	popularity_log	Log-transformed to reduce skew, then standardized
Metadata Richness	metadata_richness	Log-transformed tag counts, then standardized
Genre Indicators	18+ binary genre columns (e.g., Drama, Comedy)	One-hot encoded, no scaling needed

B. Dimensionality Reduction with PCA

The full feature space included both numeric and genre variables, which resulted in a relatively high-dimensional dataset. To reduce dimensionality while preserving important variance in the data, Principal Component Analysis (PCA) was applied. This transformation improves clustering stability and reduces the impact of redundant features.

```
pca = PCA(n_components=10)
X_pca = pca.fit_transform(X)
```

C. Cluster Algorithm Selection

Several clustering algorithms were considered, including K-Means, Hierarchical clustering, and DBSCAN.
K-Means was selected for this analysis because:

- The dataset contains dense numeric features suitable for centroid-based clustering.
- The dataset size is large, making K-Means computationally efficient since it is scalable allowing it to efficiently handle 30,000+ movies.
- Movies naturally form groups based on similar characteristics.
- The expected clusters represent general behavioral patterns rather than irregular shapes.

D. Determining the Optimal Number of Clusters

Selecting the correct number of clusters (k) is a critical step in K-Means clustering. To determine the appropriate value, the Silhouette Score was computed for cluster values ranging from k = 2 to k = 10.
The silhouette score measures how well-separated clusters are by comparing the distance between points within a cluster to points in neighboring clusters.

```
k=2, silhouette=0.937
k=3, silhouette=0.891
k=4, silhouette=0.879
k=5, silhouette=0.862
k=6, silhouette=0.836
k=7, silhouette=0.811
k=8, silhouette=0.791
k=9, silhouette=0.775
k=10, silhouette=0.764
```

While k = 2 and k = 3 generate higher silhouette scores, k = 4 was selected at the final value as it scored 0.879 which would provide a balance between cluster separation and interpretability. Using 2 to 3 clusters would only divide the data into clusters "Popular" vs. "Unpopular" providing simple and unmeaning results.

To confirm the intuition of using k = 4 provided by the silhouette score, the Elbow Method was also utilized.

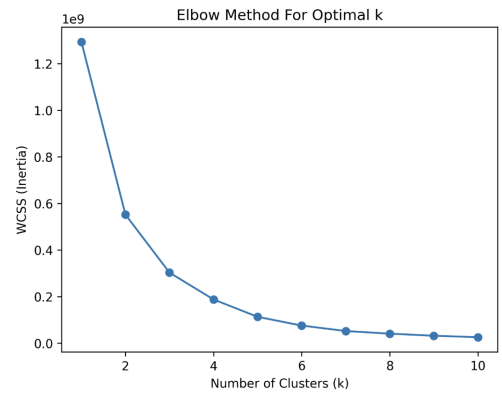

Fig. 11. Elbow Method Results

E. Visualization of Clusters

- **K-Means Clusters (PCA Projection)**: To visualize clustering results, the PCA-transformed dataset was reduced to two principal components and plotted in a scatter plot. Although PCA reduces dimensionality, distinct groupings are still visible, indicating meaningful separation between clusters. The clear separation between clusters confirms the quality of the clustering solution. Each color represents a distinct movie archetype.

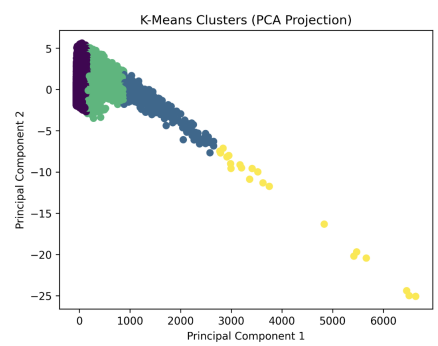

Fig. 12. PCA Cluster

- **Clusters by Popularity vs Rating**: It is evident from the following scatter plot that clusters are strongly influenced by both popularity and perceived quality, two key indicators of movie success.

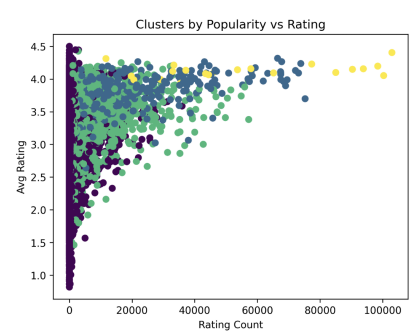

Fig. 13. Enter Caption

- **Rating Distribution by Cluster**: The graph was generated to show how average movie ratings differ between clusters revealing the distinct rating profiles of each cluster, normalized to account for the dramatic size differences between groups. Unlike a raw count histogram, density normalization allows us to compare the shape of each cluster's rating distribution regardless of cluster size.

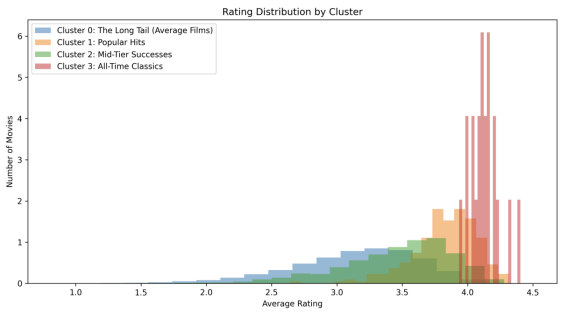

Fig. 14. Normalized Density Plot - Rating Distribution

The progressive rightward shift of peaks confirms the hierarchical nature of the discovered clusters. Each cluster occupies a distinct region of the rating spectrum, validating that K-means successfully identified meaningful divisions in how audiences rate films across different tiers of quality and popularity.

V. Findings & Interpretation

The goal of this project was to discover patterns within the MovieLens dataset by examining relationships between movie ratings, popularity, metadata richness, and genre composition. The first discovery questions originally focused on identifying whether natural groupings of movies exist and what characteristics tend to appear together.

The clustering analysis revealed four distinct groups of movies, each characterized by different combinations of popularity, rating quality, and genre distribution. These clusters indicate that movies do not exist on a single spectrum of popularity or quality, but instead fall into several recognizable categories.

The PCA visualization of clusters shows that these groupings form clear regions in the feature space, confirming that the clustering algorithm successfully separated movies based on meaningful characteristics.

```
Cluster 0 Profile:
Primary Genres: Drama, Comedy, Thriller
Avg Rating: 3.14
Avg Votes: 426
Avg Release Year: 1989

Cluster 1 Profile:
Primary Genres: Drama, Action, Sci-Fi
Avg Rating: 3.83
Avg Votes: 23742
Avg Release Year: 1997

Cluster 2 Profile:
Primary Genres: Drama, Thriller, Comedy
Avg Rating: 3.47
Avg Votes: 6915
Avg Release Year: 1990

Cluster 3 Profile:
Primary Genres: Drama, Thriller, Sci-Fi
Avg Rating: 4.13
Avg Votes: 56145
Avg Release Year: 2001
```

```
=== NATURAL MOVIE GROUPS DISCOVERED ===

 Cluster                        Name  Size          Primary Genres Avg Rating     Rating vs All Avg Votes Avg Year                                Examples
       0 The Long Tail (Average Films) 29982 Drama, Comedy, Thriller       3.14 43.1th percentile      426     1989             Babe, Clear and Present Danger
       1                  Popular Hits   258   Drama, Action, Sci-Fi       3.83 93.7th percentile    23742     1997           Jurassic Park, Schindler's List
       2             Mid-Tier Successes  1700 Drama, Thriller, Comedy       3.47 70.3th percentile     6915     1990           Independence Day, Fugitive, The
       3              All-Time Classics    21 Drama, Thriller, Sci-Fi       4.13 99.6th percentile    56145     2001 Shawshank Redemption, The, Forrest Gump
```

- **Cluster 0 - The Long Tail**: Nearly 30,000 films (93.8% of the dataset) cluster together with an average rating of 3.14 and modest viewership of 426 votes. Their primary genres, Drama, Comedy, and Thriller, represent the universal building blocks of mainstream cinema.

  This is the "background noise" of the film industry; watchable, unremarkable movies that fill out the middle of the distribution. They are not terrible, but they don't stand out either. Most films ever made live here. They are released, find a modest audience, and gradually fade from cultural memory. A business strategy could consider lowering licensing costs for this tier.Investment here should be minimal since these movies are commodities, they won't generate high popularity or long-term fans.

- **Cluster 1 - Popular Hits**: Only 258 films achieve cluster status with 3.83 average ratings (94th percentile) and nearly 24,000 votes. There is also somewhat a genre shift here as Sci-Fi is introduced suggesting that spectacle and genre ambition correlate with higher ceilings.

  Therefore, these movies are widely recognized as "good" due to their commercial success and high quality. These films drive word-of-mouth, generate repeat viewings, and attract new subscribers. For a streaming platform, these are the titles to feature prominently in marketing. They justify premium pricing and create brand association with quality. Every dollar spent acquiring or promoting these films has multiplier effects through cultural conversation.

- **Cluster 2 - Mid-Tier Successes**: This cluster consists of 1700 movies forming a distinct cluster with average ratings of 3.47 and 6,915 votes. These films have clearly broken through to find a solid audience, with ratings significantly higher than the long tail. These films represent successful mainstream releases that attract significant audiences but do not reach blockbuster-level popularity.

  This group may represent movies that are widely

watched but do not achieve the cultural impact or visibility of major hits. For content distributors, this cluster represents the "reliable performers". These films drive moderate engagement and can be used to retain casual viewers. Spending money on marketing here has predictable but limited returns.

- **Cluster 3 - All-Time Classics**: The final cluster contains only 21 movies that achieved extremely high ratings and exceptional popularity receiving 4.13 average rating and 56,145 votes. This goes to show that this group represent iconic movies that remain highly regarded by audiences over time. his cluster consistently appear in the upper end of the rating spectrum proving to have consistent rankings as the greatest of all time, with cultural resonance extending far beyond their initial release.

  From a marketing perspective, these are the "evergreen" movies. They generate engagement decade after decade, requiring no additional marketing investment. They are the foundation of permanent titles that will always have an audience. For any content platform, securing rights to these films will provide long-term, compounding returns.

A. Summary

These findings have practical implications for content strategy, marketing investment, and creative decisions. They suggest that while most films will remain in the long tail of modest attention, a small fraction can break through to achieve outsized impact. Moreover, that breakthrough is not random but follows predictable patterns of quality, genre, and cultural resonance.

VI. Limitations & Surprises

A. Limitations

- K-Means assumes clusters are spherical and evenly distributed in feature space. However, movie popularity and rating distributions are highly skewed, which may reduce the algorithm's ability to separate clusters cleanly.
- Dimensionality reduction using PCA was applied to reduce feature space complexity. While this improves computational efficiency, it may remove subtle genre distinctions that could influence clustering outcomes.
- The decision to filter for movies with at least 10 ratings was necessary for statistical stability, but it potentially limits the "Discovery" to established films.
- A fixed value of k = 4 was selected for interpretability. Although silhouette analysis and the elbow method were conducted to justify this value, different values of k may reveal alternative cluster structures.

B. Surprises

- A major surprise in the data was the Average Year for Cluster 3 (All-Time Classics) being 2001, while Cluster 0 (The Long Tail) was 1989. One might expect "Classics" to be older, but the data suggests that its status in the MovieLens ecosystem is heavily tied to the era when the platform's user base was most active, giving newer films a higher probability of accumulating the votes or tags required for Cluster 3.
- The high silhouette score for k = 2 (0.937) was surprising, as real-world data is rarely that clean. This mathematically confirms that the most dominant pattern in human movie consumption is a binary one: a film is either Popular or Unpopular.

VII. Next Steps: M3 - Complete Implementation

This report conducted a comprehensive analysis using the K-Means Clustering algorithm. However, it only addressed discovery question 1. Although, assumptions can be made about users from the outputs from movie clusters, it is still necessary to evaluate users separately.

Therefore, for the the complete implementation, discovery questions 2 and 3 will be addressed and report on user behavior and seasonal trends by continuing to use clustering techniques. Furthermore, new data mining techniques such as anomaly detection will also be utilized for deeper insights about the data and answer the remaining questions.

Finally, the project will produce comprehensive visualizations to clearly present the discovered patterns. These results will be integrated into the report, along with a discussion of insights, limitations, and potential applications of the findings on user behavior analysis.

VIII. Conclusion

This project demonstrates the power of data mining to reveal meaningful patterns in cultural data. The discovered patterns from the long tail of average films to the greatest of all-time classics reflects not just statistical artifacts but real dynamics in how audiences discover, evaluate, and remember films. In an era of abundant content and limited attention, understanding these patterns becomes increasingly valuable for creators, platforms, and consumers alike.

References

[1] "Movielens," GroupLens, https://grouplens.org/datasets/movielens/.